

Letter for publication in Science

Bates & Elman [1] provide a seriously misleading commentary on Saffran, Aslin and Newport [2]. After a two-minute exposure to strings of three-syllable nonsense "words", 8-month old infants in Saffran *et al.*'s study listened longer to stimuli that did not contain these words than they did to strings that did contain these words. Bates & Elman consider these results important because they "fly in the face of received wisdom" -- wisdom documented by three out-of-context quotations from Piatelli-Palmarini [3] and Chomsky [4]. Readers of *Science* should be alerted that the quotation from Chomsky cited as pp. 138-139 of [4] was actually created by combining a fragment of text from p.39 with another fragment from p.245 (a 206-page ellipsis marked only by brackets around the word "that"). These quotations are presented as evidence for an allegedly "widespread belief that humans cannot and do not use generalized statistical procedures to acquire language". This belief is supposedly refuted by Saffran *et al.*'s discovery that "babies can learn". In fact, the appearance of a refutation -- indeed, the appearance of a *controversy* -- arises only from Bates & Elman's inaccurate characterization of the views with which they purport to disagree. Just as their quotations confound two distinct passages from Chomsky, so their discussion confuses two distinct issues. We disentangle them below.

The first issue concerns the division of labor between innate and environmental factors in language acquisition. The existence of specific biological support for language is beyond doubt. Arguments come from the nature of structural properties common to all languages; restrictions on the degree to which languages vary; linguistic knowledge not attributable to the environment; as well as uniform patterns of normal and abnormal language development -- plus the fact that non-human mammals with good statistical learning and computational capacities [5] nevertheless do not develop language. These observations are perfectly compatible with Saffran *et al.*. No matter how rich a child's innate linguistic endowment, the fact that she acquires the language of her community tells us that she also has methods for analysing input. No "received wisdom" has ever doubted the existence of "learning" in this non-technical sense [6].

The second issue concerns the domain-specificity of cognitive functions. Some researchers have indeed questioned whether humans possess "generalized" learning mechanisms not associated with particular cognitive domains. Discussion of this topic, including the discussions in [3] and [4], uses the term "learning" in a technical sense, to refer to generalized mechanisms of this sort. Bates & Elman assert that Saffran *et al.*'s results support the existence of this "powerful" type of "learning", even calling that the "central contribution of the Saffran *et al.* report". But as Saffran *et al.* themselves noted, their study makes no such contribution. Since their work involved only linguistic stimuli, it teaches us nothing about other cognitive domains, and speaks to the question of "learning" only in the non-technical sense discussed in the preceding paragraph.

If anything, the speed with which Saffran *et al.*'s infants reacted to linguistic input hints at a possible predisposition to linguistic analysis -- a conclusion opposite to that endorsed by Bates & Elman. Unfortunately, it is not entirely clear whether Saffran *et al.*'s study involved language acquisition at all. While Saffran *et al.*'s infants listened longer to the novel stimuli, infants in other acquisition studies that use similar methods but use materials from natural languages (e.g. [7]) have shown just the opposite preference. Saffran *et al.*'s footnoted conjecture attributing the novelty effect to the familiarity of their stimuli is unlikely, given that the published experiments that produced the opposite effect used stimuli that are even more familiar. However these questions are resolved, one point is clear: the "spin" accorded to Saffran *et al.* by Bates & Elman is wholly unjustified.

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